



Integrated fuzzy and deep learning model for identification of coconut maturity without human intervention

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Abstract

Maturity of the coconut is judged manually based on the shape, color, weight, shaking sound, timeframe, etc. Currently, there is not much research being done to find solutions involving the latest techniques including fuzzy, machine learning and deep learning techniques for identifying maturity of the coconuts without human intervention. These techniques have significant challenges in detecting the multiple classes in coconut maturity due to difference in size and shape of the coconuts and the complex nature of their backgrounds as each coconut tree and each coconut bunch are unique. In this research work, we propose an integrated fuzzy and deep learning model (IFDM) to identify the maturity classes of coconut. Deep learning-based Mask R-CNN is used to classify and segment the coconut images. The combined fuzzy system (CFS) extracts the color, shape and scratch features with a fuzzy extraction technique for each feature. A probability-based fuzzy integration (PFI) method integrates the output from each fuzzy extraction to classify the coconuts. Decision-making probability (DMP) model combines the class probability of the probability-based fuzzy integration method and Mask R-CNN model to obtain the final coconut maturity class. The dataset with 10,000 images is used for training. Two test datasets, the first one containing 1754 real-time coconut images and the second containing 1000 cropped coconut images from the background, are used for testing. Using the proposed learning model in real time, an accuracy of 86.3% is obtained in classifying the maturity classes of coconuts. While integrating the models like Yolo v3, Yolo v5, Faster R-CNN, random forest and KNN with combined fuzzy system, Mask R-CNN performed better in terms of accuracy.

Keywords Deep learning · Detection of coconut maturity · Fruit grading · Fuzzy with deep learning methods · Optimal feature selection · Machine learning · Mask R-CNN.

1 Introduction

Coconuts are one of the most utilized fruits in South Asia and the Pacific Rim nations. They are used to produce oil, food, medicines, coir pith, cosmetics, etc. The Philippines,

India and Indonesia account for over 70% of global production of coconuts. Coconuts can be utilized in numerous ways depending on their stage of development. Tender coconuts are used for drinking purposes, Ayurvedic medicine and cosmetics. Matured coconuts are used for cooking. The coir from matured coconuts is used to make mats and ropes. We classify the in-between state of the tender coconut and fully mature coconut as half-matured. The half-matured coconuts are not utilized and harvested as much as tender and fully matured coconuts since they have rubber meat. Currently, the segregation of maturity classes of coconuts is carried out by visually inspecting the coconuts and/or shaking the coconuts with hands. Acute shortage of trained people to climb coconut trees and harvest coconuts is a major problem in this field [1]. There is a significant delay in harvesting coconuts which affects

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the income of the farmers. Hence, there is a strong need to develop a smart, reliable and appropriate method involving computer vision-based machine learning and deep learning techniques in classification of maturity of coconuts.

Smart agriculture has helped in continuous monitoring of the health of crops and increase in the yield. By employing various computer vision methods and image processing, machine learning (ML) and deep learning (DL) techniques, the farmers can achieve increase in crop quality. They are also able to reduce the labor costs. Accurate and efficient classification of maturity of coconuts using smart solutions poses challenges in terms of complex background of coconuts and their color, shape, size, scratch marks, etc. In addition, the sunlight can change the illumination of the coconuts and the shadows that fall on coconuts due the structure of the crown can affect accuracy of the coconut detection and classification. Recent advancements in DL have resulted in interaction between various neural networks and other artificial intelligence technologies. This has led DL to find its way into agricultural applications. In this study, a fuzzy logic and mask R-CNN-based automated coconut detection and maturity classification of three maturity classes of coconuts is proposed. The major contributions of this research work are,

- (a) Dataset creation consisting of 12,754 images of coconuts with complex background, color, shape, size and scratch marks.
- (b) Mask R-CNN-based coconut maturity classification.
- (c) Fuzzy logic-based coconut maturity classification.
- (d) Decision-making probability (DMP) model for obtaining the final coconut maturity class by combining the class probability of fuzzy logic-based method and Mask R-CNN model.
- (e) Integrating Yolo v3, Yolo v5, Faster R-CNN, random forest and KNN with combined fuzzy system, for performance evaluation.

2 Literature review

In object classification, combining machine learning models and fuzzy algorithms helps to gain better accuracy and robustness of the system. Cascading of the model became important in recent studies, suppressing the cons and increasing the overall system's pros. Some studies in the field of medicine utilized fuzzy in terms of clustering the object of interest in the image. The paper [2] used Bayesian fuzzy clustering in segmenting the tumor from the magnetic resonance imaging (MRI) image and obtained an accuracy of 98.5%. In another research [3], brain tumor segmentation using deep learning and fuzzy- K means

clustering for MRI images with 84% accuracy is proposed. Fuzzy C-mean (FCM) proposed in paper [4] achieved an accuracy of 95.7% in detecting tumor in mammograms. These methods work depending on the intensity variation of the pixel. In case of a tumor, there is a significant variation between the region with tumor and without tumor. But in the case of images of coconuts with complex background, the pixel's intensity value is scattered throughout the image and creates more non-uniform clusters. So, implementing FCM is not the appropriate method for segmenting coconut.

The paper [5] proposed a method for obtaining MRI brain images that are affected by noise and intensity in homogeneity. The existing method of using FCM does not compute the membership values of neighbor pixels, resulting in pixel misclassification. To address this, a type-2 adaptive weighted spatial FCM clustering algorithm was proposed. This algorithm would replace the fixed fuzzy value with the value of the fuzzy linguistic fuzzyfier. Studies in the field of human face recognition and plant disease detection have utilized fuzzy in classification. An approach to face detection that used a trained adaptive neuro-fuzzy inference system (ANFIS) model for classification was suggested in [6]. The accuracy of the system was increased by applying an adaptive genetic algorithm to refine the ANFIS model parameters. The paper [7] proposed a fuzzy weightage average-based fusion model offered to adjust the weightage of multimodal surface information dynamically. The paper [8] proposed a method for detecting and classifying plant diseases. These methods used in face detection and plant disease detection have their own challenges and limitations. There are multiple methods based on robotics presented in papers [9–13] which includes robotic climbers for harvesting coconuts, adding AI to robotic coconut tree climbers, Kinect-based coconut tree climber, etc., but not quite intelligent enough to detect the maturity of coconuts.

Unifying a machine learning model with fuzzy is an approach utilized to reduce system's computational power and improve its accuracy. The advantages of combining two machine learning paradigms for semantic semi-supervised learning approaches are demonstrated in paper [14]. The problem of how to create a structure that reflects data-distributed information with labeled and unlabeled data was resolved. The research [15] suggests a solution to the problem, the deep learning model encountered when classifying leaf cultivars. Author suggested deep fuzzy representation learning, merging and cascading a lot of fuzzy random forests in order to retrieve contour feature and classify the leaf. In [16], the proposed novel CoMHiSP framework based on a fuzzy support vector machine achieves an accuracy of 97.25% in classifying cancer histopathological images. Compared to the traditional

method, with low number of datasets, system performs better and robust. In all these models, accuracy improved after fusing machine learning model with their proposed fuzzy logic. But there is hardly any research in combining fuzzy logic with machine learning techniques for coconut stage classification, coconut tree diseases identification, etc.

Currently, a combination of deep learning models and fuzzy logic is gaining popularity for various applications. In [17], a combination of Mask R-CNN to localize the tomato, and the fuzzy inference system to classify the tomato achieved an accuracy of 98%. The paper [18] integrated the fuzzy and Yolo v5 for improving the object detection model for helmets and shows that the proposed model outperformed the existing model. Using Faster-RCNN and fuzzy sets to automate the detection of ripe tomatoes, which resulted in a significant reduction in mean relative error, is proposed in [19]. To show the advantage of the reduction in the number of parameters in several CNN models, integrated Yolo with convolutional fuzzy neural network is presented in the research work [20]. The deep learning models can handle unstructured data. When combined with fuzzy logic, they give better accuracy and reduce the number of network parameters used in the system.

The research [21] suggested integrating the deep learning model with the YOLO non-maximum suppression fuzzy method to increase the accuracy and speed of self-driving cars. The paper [22] classified the mangoes into unripe, half-ripe and ripe based on ripeness and good, average and poor based on quality. KNN model was utilized for quality classification, while a fuzzy classifier and CNN model were coupled for maturity classification. In addition, the integration of KNN and self-adaptive chicken swarm optimization FCNN method improved the classification accuracy. The research work presented in [23, 24] and [25] used deep learning models for classifying mango, tomato and kiwi fruit, respectively, without using fuzzy logic. But better accuracy can be achieved when combining fuzzy logic with DL algorithms.

There are extremely less research works in the literature that deals with classification of coconuts. Even those few studies in classification of coconuts use ML/DL models without fuzzy logic. In paper [26], authors used Faster R-CNN with ResNet-50 network for the coconut maturity classification as tender and matured. However, there is possibility that the half-matured coconut could fall under either tender or matured class. In addition, dataset size is very limited, and part of the dataset contains images from google, and hence, qualification of these images for proposed research work cannot be guaranteed. Another research work for coconut classification by using different CNN models is proposed in [27]. It was proved that ResNet-50 model

outperformed other CNN models. Even though authors claim to have achieved accuracy of more than 98%, only two classes of coconut maturity are defined including tender and mature. In addition, size of the dataset is not mentioned. Classification of coconuts using acoustic signal of tapping sound is proposed in [28] which uses random forest (RF), SVM and ANN models for classification of coconuts as premature, mature and over mature. However, it is a time-consuming process to tap each and every coconut to find maturity stage even though process is automated. The previous two research works [26] and [27] discuss about classification of maturity stages in pre-harvesting case, acoustic-based classification is proposed for post-harvesting.

3 Method

3.1 Proposed architecture

The proposed architecture is shown Fig. 1. The architecture flow is initialized with pre-processing and augmentation of images used to train Mask R-CNN model. The trained model can segment the coconut from the cluster of coconuts and the complex background environment. The segmented image from trained model is presented to the feature extraction and optimization algorithm. The color, shape and scratch are the features that are designated for this proposed architecture. Each feature has a different feature extraction and optimization method. The denoted features are allocated to the fuzzy system. The fuzzy system comprises of a fuzzifier, inference engine, probability-based rule and de-fuzzifier. The de-fuzzifier yields the percentage of maturity contributed by each feature set. Then, cascading of three-feature fuzzy system is done by the proposed PFI. So overall, combining a fuzzy system of each feature and PFI is proposed as combined fuzzy system (CFS). The result of the CFS and Mask R-CNN classifier are integrated with proposed DMP, which predicts the maturity of a class of coconut.

3.2 Dataset collection and pre-processing

There are several varieties of coconut trees such as Malayan Yellow Dwarf, Fiji Dwarf, Golden Malay, King Coconut, West Coast Tall, Macapuno, Green Dwarf, Panama, Chowghat Green Dwarf (CGD) and West Coast Tall (WCT). For this research work, dataset collection is made from tree varieties of CGD and WCT. The images are collected from Central Plantation Crop Research Institute (CPCRI), Kayamkulam, Amritapuri, Kerala. We started with 1,000 images collected from 96 different coconut trees. To enhance the dataset, we applied augmentation techniques, including flipping, rotating and

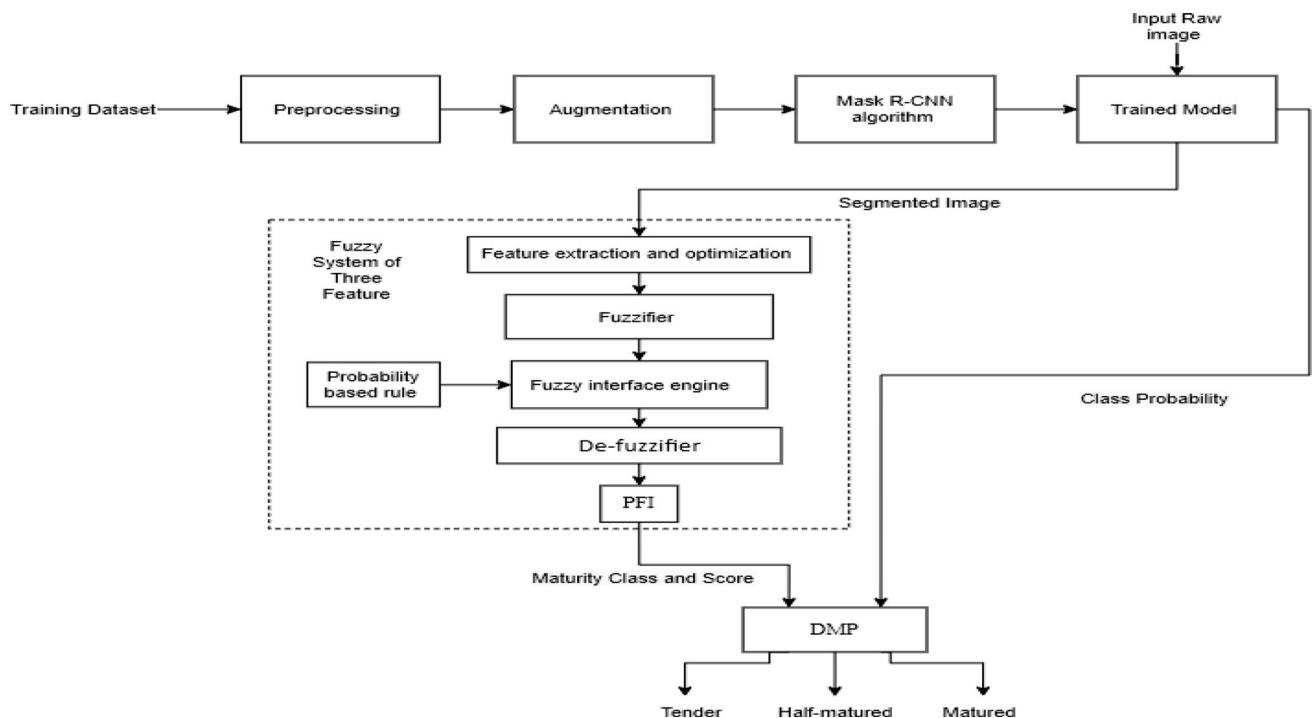


Fig. 1 Architecture diagram of the proposed IFDM

adjusting brightness, which ultimately increased the dataset to a total of 10,000 images. For model training, 80% of the data (8000 images) was employed, while 2000 images (20%) were reserved for validation. In addition, two separate datasets were created, one consisting of 1,754 coconut images and another comprising 1000 cropped coconut images extracted from the background, as detailed in Sect. 4.1. In total we created a dataset of 12,754 images.

The three maturity classes of coconut considered in this research work are tender, half-matured and mature coconuts. Each class has a different period of growth: for tender coconut it is 7–8 months; for half-matured coconut it is 9–11 months; and for matured it is 12 months. The color, shape and scratch marks are the visible distinctive factors in categorizing the coconut. These three factors form the feature set used to differentiate the three maturity classes of coconut. Tender coconuts of CGD and WCT varieties are mostly green in color and round with minimal scratch marks. In case of half-matured coconut, the color is an average mix of brown and green and the shape transitions from round to oval with average scratch marks. Matured coconuts are brown in color, ovoid shaped, with deep scratch marks. With these features, three maturity classes of the coconut dataset were collected.

3.3 Mask R-CNN

Mask R-CNN algorithm is used for segmenting the object of interest in the image from the complex background.

There are alternative methods to Mask R-CNN existing with less computational complexity like fuzzy C-means and K-means clustering. Such methods fail in the segmentation of coconut from complex environments as they depend upon a lot of constant parameters. The parameters vary when the background changes significantly. The other alternative methods are Yolo and Faster R-CNN. These methods can localize the coconuts for classification, but localization is not much accurate particularly when multiple coconut bunches are in a single image. The obtained bounding box from Yolo and Faster R-CNN includes the coconut and part of the background image. This background image affects the efficiency of feature extraction algorithms. Mask R-CNN can classify and localize the coconuts from the background by using polygonal bounding points.

Figures 2a and 3a are the sample test images passed to the trained model and Figs. 2b and 3b represent the output of the Mask R-CNN model. Figures 2c and 3c show the cropped image from complex environments. In addition to the coconut segmentation, the Mask R-CNN is used to classify the coconuts into tender, half-matured and matured. The model is trained with images of tender coconut, half-matured and matured coconuts with the label. The trained model provides class score of each image and that can be represented as A_j , where $j = (1, 2, 3)$. Each j value represents score of each class. Value of A_j considers a Gaussian distribution where $0 \leq A_j \leq 1$.

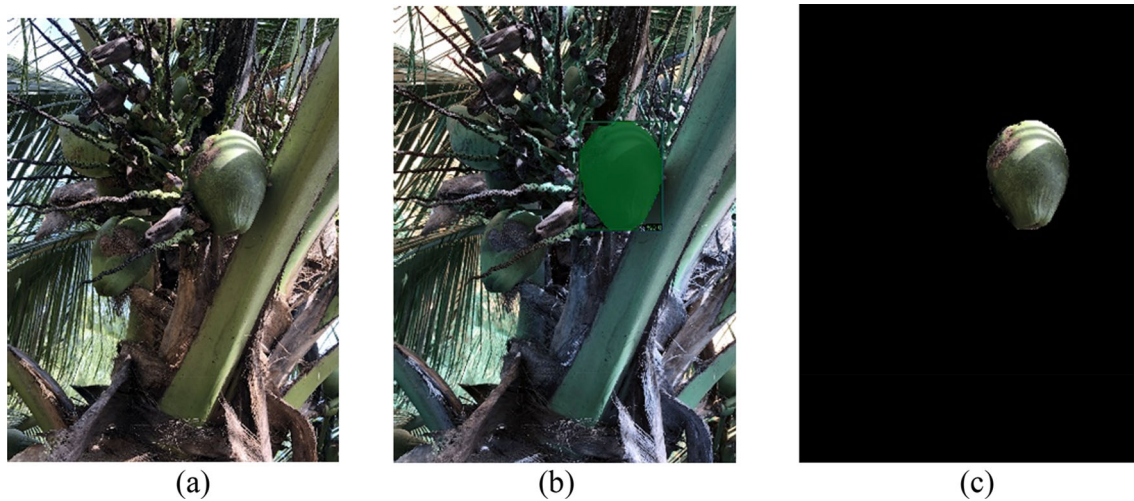


Fig. 2 Mask R-CNN segmentation process for single coconut: (a) Input image (b) localization of object of interest (c) segmented coconut from background environment

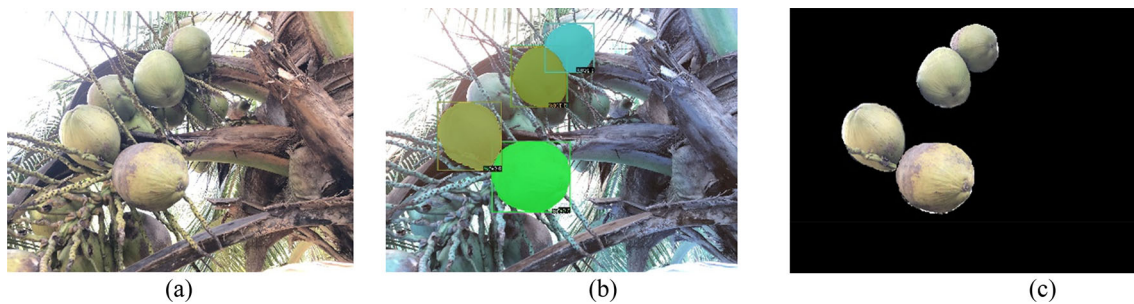


Fig. 3 Mask R-CNN segmentation process for multiple coconuts: (a) Input image (b) localization of object of interest (c) segmented coconut from the background environment

3.4 Fuzzy model

The fuzzy model comprises of fuzzifier, an inference system and a de-fuzzifier. The fuzzifier is used to convert the value obtained from feature extraction to fuzzy values. The fuzzy value transition depends upon the membership function (μ) used in the system. The fuzzifier converts the input signal to the percentage of poor, average and good which represent the level of maturity of the coconut. Each percentage is mapped to the membership function (MF) value and combined with the final output from the fuzzifier. The output from the fuzzifier is subjected to an inference system. The inference system produces output depending on the IF–THEN rule described later in this section. Further, this output is passed to the de-fuzzifier which converts the aggregated output from the inference engine to a crisp value with respect to the de-fuzzifier MF. Centroid-based method for defuzzification is used to obtain equal probability in the classification of coconuts. The accuracy

determination of class of the maturity of the coconut (tender, half-mature, mature) depends upon the membership function and rules proposed. For the distinct visible difference between coconut classes, the chosen feature set of the proposed work are color, shape and scratch.

3.4.1 Color feature extraction algorithm and membership function

Mask R-CNN trained model crops the coconut part from the image of the tree crown. The segmented image is subjected to the color extraction algorithm. In image classification, the coconut images are widely affected by sunlight and shadow. To account such pixel explosion in image, the color extraction is used to convert the segmented image into Hue Saturation Value (HSV) color space. A binary mask of green and brown is created to obtain the percentage of green and brown in coconut instead of the whole image.

Algorithm 1 Feature extraction-color

Step1: Input the Image.
 Step2: Convert to HSV color space.
 Step3: Obtain the number of pixels contributed by coconut in the image.
 Step4: Set the NumPy array maximum and minimum for green and brown.
 Step5: Use bitwise operation to obtain the number of pixels of green and brown.
 Step6: Obtain the percentage of a green and brown pixel by dividing the number of respective color pixel by total number of pixel accounts coconut in the image.
 Step7: Output the percentage of green and brown present in the coconut.

From Algorithm 1, percentage of green and brown color are obtained which is passed to fuzzifier. Figure 4a represents the MF for green (which is same for brown) which is used for fuzzification. Both MF (for green and brown) are trapezoidal with same start, stop and step value. The choice of type of MF depends on computational power of hardware, and number of breakpoints in a feature set needed to classify each class. Fuzzifier outputs the cluster of values to inference engine. Color inference engine outputs fuzzy value depending on rule mentioned below.

Rule 1 If green is good and brown is poor then classify as Tender.

Rule 2 If the brown is average and green is average then classify as Half-matured.

Rule 3 If green is poor and brown is good then classify as Matured.

From the inference system, the aggregated fuzzy value is obtained and subjected to the de-fuzzifier. Figure 4b represents the MF of the de-fuzzifier which converts the

fuzzy value to crisp output. This crisp output from fuzzy system for color feature represents the maturity of coconut depending only on percentage of green and brown color from image.

3.4.2 Shape feature extraction algorithm and membership function

For shape, the proposed extraction algorithm determines the percentage of closeness of the coconut being round (deformation). The tender coconut mostly is circular or oval. Initially, the input image is converted to a gray image to find the contour. The segmented images from Masked R-CNN gives the perfect contour for each coconut. Algorithm 2 represents the technique of extracting the shape feature. The output from the feature extraction block would be percentage of deformation in the coconut.

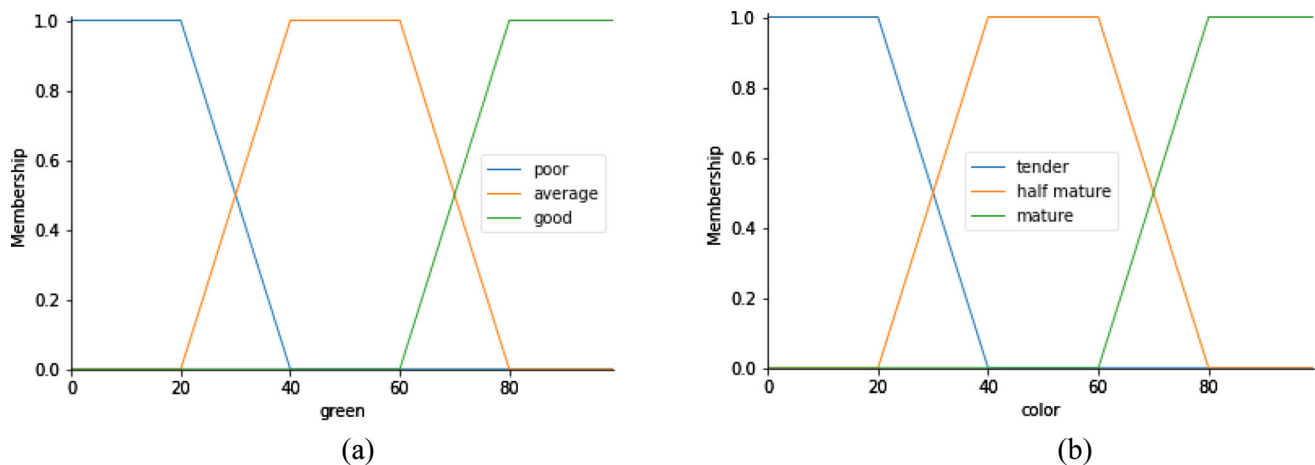


Fig. 4 (a) Color MF of fuzzifier (b) Color MF of de-fuzzifier

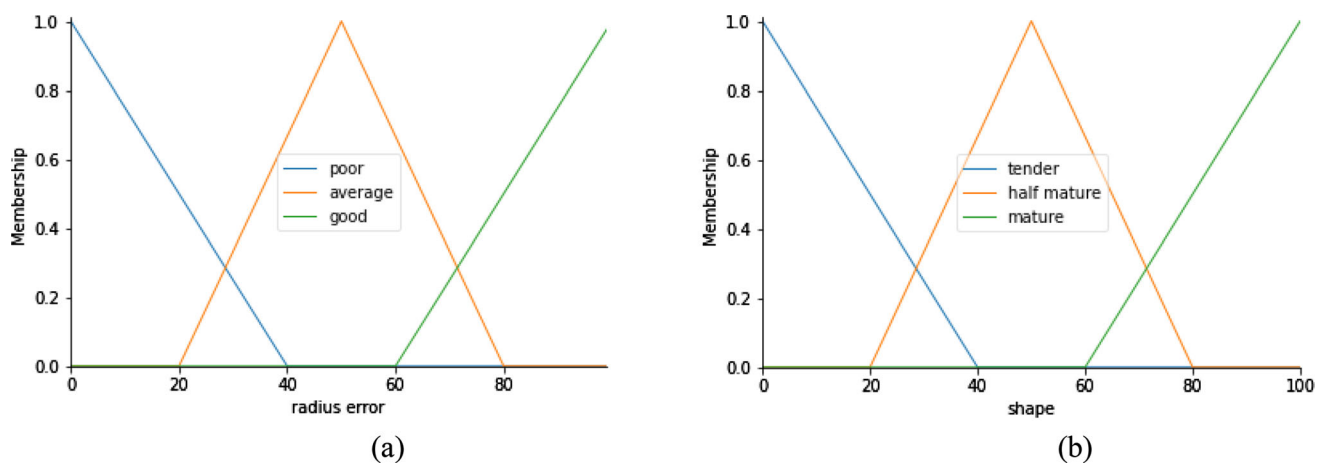


Fig. 5 (a) Shape MF of fuzzifier (b) Shape MF of de-fuzzifier

Algorithm 2 Feature extraction-shape

Step1: Input the Image.
 Step2: Convert to grey image color space.
 Step3: Find binary thresholding of the grey image.
 Step4: Find the contours of an image and obtain the contour with the highest data points.
 Step5: Obtain the centroid of the selected contour.
 Step6: Find the distance between contour points to the calculated centroid.
 Step7: Obtain the area and perimeter of the selected contour.
 Step8: Calculate the difference between the circle and the selected contour.
 Step9: Output the deformation of structure.

The feature extraction method, MF and rules implemented are varying. In fuzzifier for shape, there is a single input obtained from the feature extraction block. The MF used here is triangle MF, which provides a better result compared to trapezoidal and sigmoidal MF with simple computation power. Figure 5a represents the fuzzifier MF for the shape. The rules of the inference engine are mentioned below:

- Rule 1* If the percentage of radius error is poor then classify as a Tender.
- Rule 2* If the percentage of radius error is average then classify as Half-matured.
- Rule 3* If the percentage of radius error is good then classify as Matured.

The cluster of fuzzy values output from the inference engine is passed to the de-fuzzifier. The breakpoint is the same for the fuzzifier and de-fuzzifier. The crisp output obtained from de-fuzzifier represents the maturity of coconut depending upon the shape.

3.4.3 Scratch feature extraction algorithm and membership function

The scratch feature extraction is complex compared to color and shape feature due to the presence of scratch in every maturity class of coconut. In the tender coconut too the presence of scratches can be observed due to the rubbing of neighbor coconut. The feature extraction method should be capable of differentiating the scratches due to dashing or aging. Algorithm 3 is used for scratch feature extraction. Sobel and K-means are used to obtain the deep scratch marks and shrinkage marks of coconut from the images. Sobel helps to detect the edges that have a sudden change of derivative value that is greater than threshold values. K-Means is used to cluster the gray region of different levels of intensities. The constant threshold is obtained through trial-and-error method. The region above the threshold is considered as scratches due to aging. The output of this extraction method would be a level of gray due to aging.

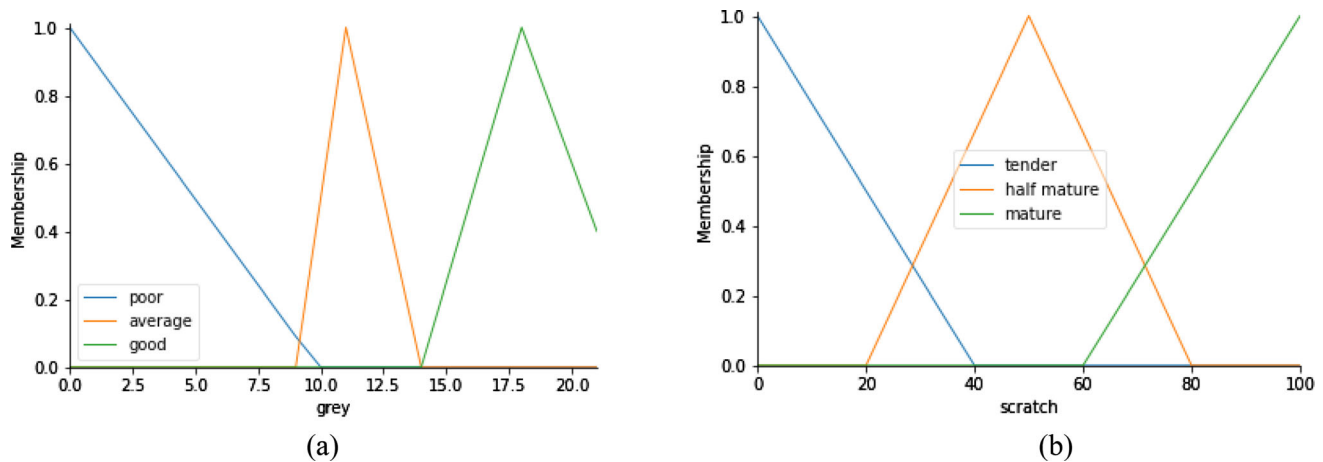


Fig. 6 (a) Scratch MF of fuzzifier (b) Scratch MF of de-fuzzifier

Algorithm 3 Feature extraction-scratch feature

Step1: Input the Image.
 Step2: Convert image to BGR color space.
 Step3: Apply Sobel edge detection on the image of the x-axis.
 Step4: Do K-Means to the Sobel edge detected image and obtain K-Means clustered image.
 Step5: Segregate the color in the image.
 Step6: Calculate the level of a grey part in the image that corresponds to the scratch in the image.
 Step7: Output the quantity of grey inside the coconut image.

Here, the fuzzifier MF is represented in Fig. 6a, which shows that the breakpoint does not look linear. The breakpoint is chosen through the trial-and-error method which provides the optimum result. The intersection between MF is also too less or none. The rules of the inference engine for the scratch are defined below:

Rule 1 If the effective gray is poor then classify it as a Tender.

Rule 2 If the effective gray is average then classify it as Half-Matured.

Rule 3 If the effective gray is good then classify it as Matured.

The defuzzification MF in Fig. 6b converts the cluster of fuzzy values to crisp output. The output represents the maturity level of coconut due to the scratch feature.

3.5 Probability-based fuzzy integration (PFI) method

The combined fuzzy system (CFS) includes the PFI method of integrating the fuzzy system of each feature. This integration mainly depends on the percentage of each feature correctly predicting the class of the feature. The proposed method starts with the collection of the dataset of 1754 images. The images chosen are not augmented but cropped

from a complex background environment. The real-time coconut images are used and manually classified in their class.

$F(c)$ –Percentage of the maturity level of coconut by color.

$F(s)$ –Percentage of the maturity level of coconut by shape.

$F(m)$ –Percentage of the maturity level of coconut by Scratch.

$P(c)$ –Probability that color decides maturity level correctly.

$P(s)$ –Probability that shape decides maturity level correctly.

$P(m)$ –Probability that scratch decides maturity level correctly.

$F(c)$, $F(s)$ and $F(m)$ are output from the de-fuzzier of each feature fuzzy system. The required output parameters of the proposed method are $P(c)$, $P(s)$ and $P(m)$. The output parameters from PFI are kept constant. It shows the importance of each feature set contributing to the proposed architecture. n_1 , n_2 and n_3 represent the number of coconuts detected correctly by the fuzzy system for color, shape and scratch, respectively. n_4 , n_5 and n_6 represent the number of coconuts detected correctly by pairing the fuzzy system where n_4 is combination of fuzzy system for color and scratch, n_5 is the combination for shape and scratch, n_6

is the combination for scratch and color. $n7$ represents the number of coconuts detected correctly by combining all three features.

$$P(c) = \frac{n1 + \frac{n4}{2} + \frac{n6}{2} + \frac{n7}{3}}{N} \quad (1)$$

$$P(s) = \frac{n2 + \frac{n4}{2} + \frac{n5}{2} + \frac{n7}{3}}{N} \quad (2)$$

$$P(m) = \frac{n3 + \frac{n5}{2} + \frac{n6}{2} + \frac{n7}{3}}{N} \quad (3)$$

$$\text{The combination of } P(c) + P(s) + P(m) = \frac{n1+n2+n3+n4+n5+n6+(3*n7)}{N} = 1$$

After obtaining the $P(c)$, $P(s)$ and $P(m)$, the $F(c)$, $F(s)$ and $F(m)$ of the test image can be directly integrated with the obtained probability value as shown in (4).

$$F = P(c) * F(c) + P(s) * F(s) + P(m) * F(m) \quad (4)$$

The fuzzy integrated ‘ F ’ value represents the percentage of maturity level of coconut. The boundary limits ‘ a ’ and ‘ b ’ are obtained from the statistical analysis of the data collected. For obtaining the boundary limits, we have collected 500 cropped coconut images with known maturity level. ‘ F ’ value is compared with the known maturity level to obtain the limits ‘ a ’ and ‘ b ’ that would have the maximum accuracy.

The output from the fuzzy PFI:

$0 \leq F \leq a$ classified as Tender.

$a \leq F \leq b$ classified as Half-matured.

$b \leq F \leq 100$ classified as Matured.

The output from the PFI is mapped from 0 to 100 using the give equation to obtain the score for each class.

$B_i = (F - a) * \frac{(100)}{b-a}$, $0 \leq B_i \leq 1$ where B_i represents score of each class.

3.6 Decision making probability (DMP) model

The outputs from the CFS and the deep learning model are given as input to the DMP model. The first input is from the CFS, and the second input is from the deep learning model. Considering A_j and B_i as illustrated in the previous sections, we define,

A_1, B_1 —classified score of Tender.

A_2, B_2 —classified score of Half-matured.

A_3, B_3 —classified score of Matured.

Following the acquisition of i, j, A_j and B_i , the following cases are defined for various combinations of i, j, A_j and B_i .

Case 1 Suppose the class prediction of CFS and the deep learning model are the same i.e., $i = j$. Then, the output of the overall system is the same class. It represents the

logical AND operation between the prediction of CFS and the deep learning model.

Case 2 If the class from CFS and the deep learning model are not the same, then probability of having the result from fuzzy correct $P(f)$ and the probability of having the result from deep learning correct $P(d)$ have to be determined. Equations (5) and (6) represent the formulation of $P(f)$ and $P(d)$, respectively. Here, ‘ C ’ represents the highest score between the B_i and A_j .

$$P(f) = C/A_j \quad (5)$$

$$P(d) = C/B_i \quad (6)$$

Case 2(1) If $P(f) > P(d)$, then the class determined by CFS can be determined as output.

Case 2(2) If $P(f) < P(d)$, then the deep learning class can be determined as output.

Case 2(3) If $P(f) = P(d)$, then the closeness of the score to the median value is the deciding factor. If the deep learning score exceeds 0.75, then the class from the deep learning model is considered as output. If the deep learning score is equal to or less than 0.75, the class received from the fuzzy probability model is considered as output.

4 Experiment and result

4.1 Test setup

Two types of test datasets are used to validate the IFS proposed in this work. These two types are created from the 200 full images of the tree crowns. Since the Mask R-CNN is capable of polygonal segmentation, the test dataset A, which consists of 1754 individual coconut images with complicated backdrop surroundings, is utilized only to evaluate the Mask R-CNN. The test images of dataset A are cropped by Mask R-CNN and subjected to the proposed CFS and decision-making model for the evaluation. 1000 background-free images of individual coconuts which form

Table 1 Evaluation of CFS block

Methods	Accuracy (%)
Equal weightage (c:m:s = 1:1:1)	72
Proportionate probability A. (c:m:s = 2:1:1)	76.7
Proportionate probability B. (c:m:s = 1:2:1)	55.9
Proportionate probability C. (c:m:s = 1:1:2)	64.7
Proposed PFI model	82.7

Table 2 Evaluation of DMP method

Methods	Precision (%)	Accuracy (%)	Recall (%)	F1-score (%)
AND	63.7	63.7	100	38.9
Fixed ratio method	76.5	76.5	100	43.3
DMP method	86.3	86.3	100	46.3

the test dataset B are cropped and used to assess other DL models to compare with the performance of the Mask R-CNN model. In addition, the test dataset B is also used to validate the proposed fuzzy, PFI and probability-based decision-making model. The intersection over union (IOU) used for testing all DL models is 0.5. The system configuration used for testing is 11th gen Intel i7, 16 GB RAM, Nvidia RTX A4000 graphics card and Nvidia GeForce RTX 3070 graphics card.

4.2 Analysis of CFS block

The test dataset B is utilized in this standalone analysis of the CFS block. The outcome of the PFI model is compared with the equal proportional weightage technique and the 2:1:1 weightage technique. In the equal weightage technique, the de-fuzzifier output from each class is given equal importance. In the 2:1:1 weightage technique, the output from the de-fuzzifier from each class is given importance as the ratio of 2:1:1. $F(c)$, $F(s)$ and $F(m)$ represent the fuzzy system output of color (c), scratch (s) and shape (m), respectively. Table 1 shows that the proposed PFI model gives better accuracy than other methods. Moreover, we can infer that color is valued more compared to other features as the second highest accuracy in Table 1 is given more weightage to color. Predominantly, the color feature part plays a significant role in classification.

4.3 DMP analysis

Dataset A is used in the analysis of DMP model. The Mask R-CNN plays a role in cropping out the coconuts from the images with complex background. The DMP method is validated by two other methods. The first one is by introducing AND logic to combine DL model and CFS. The second method of validation is by using the fixed ratio method. This method integrates the DL model and CFS by using the DL class prediction score. If the score is above 0.75, then prediction from the DL model is considered the

final output class; otherwise, output of CFS is regarded as the final output class. Table 2 shows comparison between AND, fixed ratio and DMP model.

Table 2 shows that the proposed DMP model gives greater accuracy, precision and F1-score compared to the other two methods.

4.4 CFS fuzzy classifier analysis

The fuzzy classifier used in CFS is evaluated by comparing the fuzzy classifiers of [17, 22]. In fuzzy classifier [17], color feature is used for classification. Feature extraction is done by masking the image with the upper and lower range of each color using MF. Rules for defining fuzziness as per the fuzzy classifier [17] are:

Rule 1 If green is average AND yellow is poor AND orange is average AND red is average, then classify as Tender.

Rule 2 If green is good AND yellow is average AND orange is average AND red is average, then classify as Half-Matured.

Rule 3 If green is good AND yellow is average AND orange is poor AND red is poor, then classify as Matured.

In case of fuzzy classifier [22], the image is converted to HSV color space. The green color is utilized as a feature set for this classifier. Rules for defining fuzziness as per the fuzzy classifier [22] are,

Rule 1 If green is good, then classify as Tender.

Rule 2 If green is average, then classify as Half-matured.

Rule 3 If green is poor, then classify as Matured.

Triangular MF is used in fuzzy classifier [17] and [22]. The test dataset A is used for comparison. Initially, images are input into Mask R-CNN model for obtaining cropped images of coconut. Cropped images are classified with fuzzy classifiers [17] and [22], and CFS fuzzy classifier.

Table 3 Evaluation of CFS fuzzy classifier

Models	Precision (%)	Accuracy (%)	Recall (%)	F1-score (%)
Fuzzy classifier. [17]	44.2	43.8	97.7	30.4
Fuzzy classifier. [22]	73.5	72.8	98.6	42.1
CFS fuzzy classifier	74	74	100	42.5

Evaluation of Integrated of Deep learning and CFS

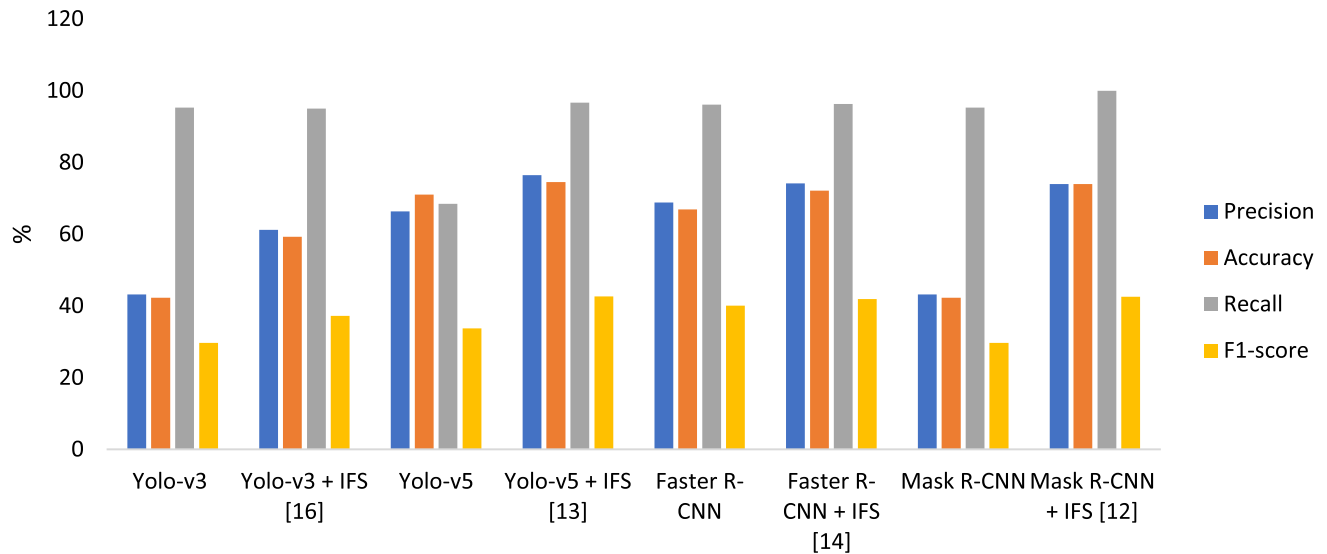


Fig. 7 Evaluation of Integrated of Deep learning and CFS

Table 4 Evaluation of integrated ML and CFS

Models	Accuracy (%)	F1-score (%)
SVM. [29]	73.3	42.7
SVM [29] + CFS	77.4	45.5
RF. [30]	50	32.9
RF [30] + CFS	58.8	37.1
KNN. [31]	66.2	38.3
KNN [31] + CFS	71	41.5

Table 3 shows that the proposed CFS fuzzy classifier is better than fuzzy classifier [17] and [22].

4.5 Evaluation of CFS with various DL models

Standard deep learning models are combined with the proposed CFS to evaluate the performance of the CFS to identify which deep learning algorithm is performing better. Yolo v3, Yolo v5, Faster-R-CNN and Mask R-CNN are considered for the evaluation. Dataset B is used for evaluation except Mask R-CNN. All the models are trained using 10,000 datasets of cropped coconut images.

According to Fig. 7, the performance of the deep learning models considered is improved after integration with the CFS in terms of accuracy, precision, recall and F1-score. Mask-RCNN + CFS combination is giving the better result compared to other combinations. If we use Mask-RCNN alone the precision, accuracy, recall and F1-score are 43.2, 42.3, 95.3 and 29.7, respectively, whereas we got 74, 74, 100 and 42.5 for precision, accuracy, recall and F1-score when Mask-RCNN + CFS combination is

used. Yolo V5 + CFS gives a comparable performance of Mask R-CNN + CFS.

4.6 Standard evaluation of CFS with different ML algorithms

In this section, the machine learning models, support vector machine (SVM), K-nearest neighbors (KNN) and RF, are integrated to verify the robustness of the proposed logic of CFS. As this is evaluation of CFS alone, we did not consider the segmentation part using Mask-RCNN. The machine learning models were trained with a 10,000 cropped coconut images dataset, and test dataset B was utilized for evaluation. The ML model classifies the coconut images with the outputs as the image's score and class. Then, models are integrated with the proposed fuzzy logic. The result of the comparison of various ML models is shown in Table 4.

Table 4 concludes that among the ML models without CFS, SVM model performs better than the RF and KNN. It also shows that the accuracy and F1-score increase when these ML models are integrated with the CFS. The accuracy increased by 8.8, 4.8 and 4.1% for RF, KNN and SVM, respectively. Similarly, F1-score increased by 4.2, 3.2 and 2.8% for RF, KNN and SVM, respectively, when the model is combined with the CFS.

4.7 Evaluation of the IFDM architecture with standard DL models

In this section, we are comparing the IFDM architecture with other DL models integrated with CFS and DMP. The

Evaluation of DL with Proposed Architecture

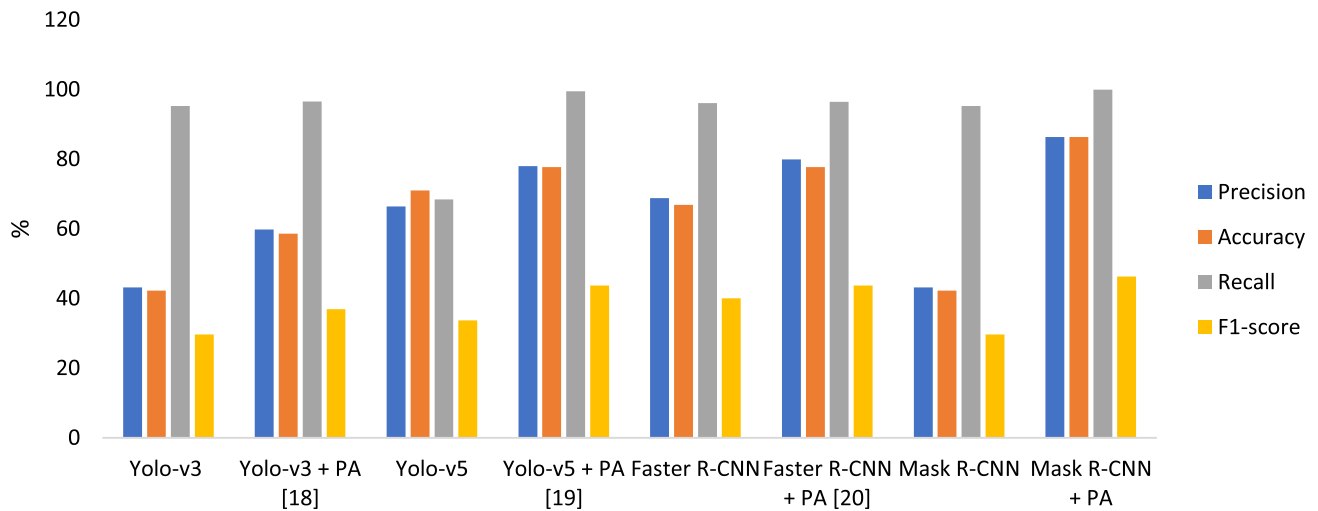


Fig. 8 Evaluation of DL with Proposed Architecture

DL models, Yolo v3, Yolo v5 and Faster R-CNN, are used for comparison. The evaluation is performed using the dataset B. The proposed architecture is evaluated using dataset A. The results obtained are shown in Fig. 8.

Figure 8 shows that the proposed IFDM architecture gives better performance in terms of the accuracy, precision and F1-score. Yolo v3 [23] + CFS + DMP is having least accuracy of 59.8% compared to other models. The Yolo v5 [24] + CFS + DMP, Faster R-CNN [25] + CFS + DMP are giving the accuracy of 78.1 and 79.9%, respectively. The proposed architecture is giving the highest accuracy of 86.4% among other methods.

5 Conclusion

In this article, we presented an integrated fuzzy and deep learning model (IFDM) for coconut maturity detection by fusing a deep learning model with a fuzzy system. There are three essential components to the proposed system: Mask R-CNN, segmented images and prediction scores for the relevant class. The segmented image from Mask R-CNN passed to CFS. The proposed CFS is modeled by combining the fuzzy system of each feature by PFI. The DMP method integrates the output of Mask R-CNN and CFS. The IFDM can localize the coconut and classify it into three classes including tender, half-matured and matured. A test dataset of 1754 images used to verify the proposed architecture demonstrated precision and accuracy of 86.3% and a recall rate of 100%. The proposed logic is then combined with various machine learning models to assess the robustness of CFS and the decision-making technique. The results demonstrated that applying the

combined method increased the accuracy of each machine learning model. Future works will emphasize the robustness of the proposed architecture on more variety of palm trees. Currently, the proposed system is validated with only two varieties of palm trees.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article. The authors have no relevant financial or non-financial interests to disclose.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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